Import Data using Transform Map: Automated Data Loading in ServiceNow

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Team ID	SWTID-2026-2169
Project Name	Import Data using Transform Map
Maximum Marks	4 Marks

Project Description:

This project demonstrates how bulk data stored in an external spreadsheet can be loaded into a ServiceNow instance reliably, without manual data entry. Data is uploaded from an Excel file through a **Data Source** into a temporary **Import Set (staging) table**. A **Transform Map** then maps each staging column to the matching field of the target table, applies a **coalesce** rule so that existing records are updated instead of duplicated, and inserts the remaining rows as new records.

The solution covers creating the target table, defining the data source, generating the import set, building the transform map with field maps and an optional transform script, running the transform, and verifying the results through the Transform History and the target table list. This approach reduces errors, saves time and keeps data consistent when migrating or refreshing large data sets.

Scenarios

Scenario 1: An HR administrator receives an Excel sheet of 500 new employees. Instead of creating records one by one, the administrator uploads the file as a data source, loads it into an import set and runs the transform map. All 500 records appear in the Employee Records table within minutes.

Scenario 2: An updated sheet contains some employees who already exist, with changed department or email values. Because Employee ID is set as the coalesce field, the transform updates the existing records and inserts only the genuinely new ones, avoiding duplicates.

Scenario 3: Some rows have inconsistent data (for example lower-case department names or a blank email). A transform script normalises the values, and rows that fail are reported as ignored or skipped in the Transform History so they can be corrected and re-imported.

Technical Architecture:

Excel / CSV file	Data Source (sys_data_source)	Import Set Table (u_import_employee)	Transform Map + Field Maps + Coalesce	Target Table (u_employee)

Description: The solution follows the standard ServiceNow import architecture. The source file is attached to a Data Source record. Loading the data creates an Import Set, whose rows are stored in a temporary staging table that mirrors the spreadsheet columns. The Transform Map reads each staging row, converts and maps the values to the target table, and either inserts a new record or updates a matching one according to the coalesce field. Results are logged in the Import Set Rows and Transform History for auditing.

Components

Component	Purpose
Data Source	Defines the file type (Excel), sheet number, header row and the attached file.
Import Set Table	Staging table created automatically with one column per spreadsheet column.
Transform Map	Links the staging table (source) to the target table and controls how rows are processed.
Field Maps	Map each source column to a target field (auto-map or manual).
Coalesce	Field(s) used to detect an existing record; matching rows are updated, others inserted.
Transform Script	Optional onBefore/onAfter script for cleansing, defaults or validation.

(Note: If you are using a database in your project you need to add an ER Diagram along with a description. Here the data model is a single staging table mapped to a single target table.)

Pre-requisites:

- **ServiceNow Personal Developer Instance (PDI):** free instance from developer.servicenow.com with admin access.
- **ServiceNow Fundamentals:** navigating the Application Navigator, lists, forms and tables.
- **Table and Field Concepts:** creating custom tables, columns and field types.
- **Import Sets Concepts:** data sources, import set tables and transform maps.
- **Excel / CSV Basics:** preparing a clean file with a header row.
- **Basic JavaScript:** for writing simple transform scripts (optional).

Project Workflow:

Milestone 1: Instance Setup and Data Preparation

- **Activity 1.1:** Sign up for and log in to a ServiceNow Personal Developer Instance.
- **Activity 1.2:** Prepare the Excel file with a header row and sample records.

Milestone 2: Target Table Creation

- **Activity 2.1:** Create the target table Employee Records (u_employee).
- **Activity 2.2:** Add the required fields (Employee ID, Name, Email, Department, Location).

Milestone 3: Data Source and Import Set

- **Activity 3.1:** Create a Data Source and attach the Excel file.
- **Activity 3.2:** Load the data to create the import set (staging) table.

Milestone 4: Transform Map Development

- **Activity 4.1:** Create the Transform Map from the staging table to the target table.
- **Activity 4.2:** Create field maps and set the coalesce field.
- **Activity 4.3:** Add an optional transform script for data cleansing.

Milestone 5: Run Transform and Verification

- **Activity 5.1:** Run the transform and review the Transform History.
- **Activity 5.2:** Verify the records in the target table and test the update behaviour.

Milestone 6: Conclusion

Milestone 1: Instance Setup and Data Preparation

This milestone prepares the ServiceNow environment and the source data that will be imported.

Activity 1.1: Get and Log in to a ServiceNow Instance

- **Step 1:** Visit https://developer.servicenow.com and sign up for a free developer account.
- **Step 2:** Click **Request Instance** and choose the latest release. Note the instance URL, user name (admin) and password.
- **Step 3:** Open the instance URL, log in as admin, and confirm the Application Navigator is visible.

Activity 1.2: Prepare the Excel Data File

Create an Excel file (Employee_Data.xlsx) with the header in the first row and one record per row:

Employee ID	Name	Email	Department	Location
E1001	Anita Rao	anita.rao@example.com	HR	Chennai
E1002	Rahul Mehta	rahul.mehta@example.com	IT	Bengaluru
E1003	Sara Khan	sara.khan@example.com	Finance	Hyderabad

Keep column names simple, avoid blank rows and merged cells, and ensure Employee ID values are unique.

Milestone 2: Target Table Creation

Activity 2.1: Create the Table

- Navigate to **System Definition > Tables** and click **New**.
- Set **Label** to Employee Records; the name is generated as u_employee. Tick **Create module** and **Create mobile module** if required, then click **Submit**.

Activity 2.2: Add the Fields

Column label	Type	Notes
Employee ID	String	Unique; used as coalesce field
Name	String	Full name
Email	String	Email address
Department	String	Department name
Location	String	Work location

Milestone 3: Data Source and Import Set

Activity 3.1: Create the Data Source

- Navigate to **System Import Sets > Administration > Data Sources** and click **New**.

- Enter **Name** (Employee Data Source), set **Import set table label** (Import Employee) and confirm the table name (u_import_employee).
- Set **Type** to File, **Format** to Excel (.xlsx), **File retrieval method** to Attachment, **Sheet number** to 1 and **Header row** to 1.
- Save the record, then use the paperclip icon to attach Employee_Data.xlsx.

Activity 3.2: Load the Data

- On the data source form, click **Test Load 20 Records** to preview the data, then click **Load All Records**.
- ServiceNow creates an Import Set and populates the staging table with one row per spreadsheet record.
- Open **System Import Sets > Import Sets** to confirm the state is **Loaded** and check the row count.

Milestone 4: Transform Map Development

Activity 4.1: Create the Transform Map

- From the import set, click the **Loads Data** related link or go to **System Import Sets > Administration > Transform Maps** and click **New**.
- Set **Name** to Employee Import Map, **Source table** to Import Employee (u_import_employee) and **Target table** to Employee Records (u_employee).
- Tick **Run business rules** if required and click **Submit**.

Activity 4.2: Field Maps and Coalesce

- Open the transform map and, in the **Field Maps** related list, click **Auto Map Matching Fields**. Review and fix any unmatched fields manually.
- Open the Employee ID field map and tick **Coalesce**, so existing records are matched on Employee ID and updated instead of duplicated.

Source field	Target field	Coalesce
u_employee_id	Employee ID	Yes
u_name	Name	No
u_email	Email	No
u_department	Department	No
u_location	Location	No

Activity 4.3: Transform Script (Optional)

Add an **onBefore** script to normalise values and skip incomplete rows:

```
(function runTransformScript(source, map, log, target) {     // Tidy
department name     target.u_department =
source.u_department.toString().trim().toUpperCase();     // Skip rows without
an email     if (source.u_email.nil()) {         ignore = true;
log.warn('Skipped row - missing email: ' + source.u_employee_id);     }
})(source, map, log, target);
```

Milestone 5: Run Transform and Verification

Activity 5.1: Run the Transform

- Open the import set and click **Transform** (or use the transform map related link) and choose Employee Import Map.
- Wait for the state to change to **Complete**. Open **System Import Sets > Transform History** to see the counts of inserted, updated, ignored and skipped rows.
- Review the Import Set Rows list; each row shows its result (inserted, updated, ignored or error) and a comment.

Activity 5.2: Verify the Results

- Navigate to **u_employee.list** and confirm that all valid records are present with correct values.
- **Update test:** change a value (e.g. Department) in the Excel file, load and transform again, and confirm the existing record is updated while the record count stays the same.
- Correct any rows that were ignored or skipped, then re-import them.

Important Notes:

- Always test with **Test Load 20 Records** before loading a large file.
- A coalesce field must uniquely identify a record; otherwise duplicates or wrong updates may occur.
- Imports should be tested in a sub-production instance before running on production data.

Exploring the Application Pages:

Page / Module	Description
Data Sources	Lists file-based sources; shows the attachment, format and load options.
Import Sets	Shows each load with its state, row count and target table.
Transform Maps	Lists maps; the form contains field maps, coalesce settings and scripts.
Transform History	Summary of each transform run with inserted, updated and ignored counts.
Employee Records (u_employee)	Final target table showing the imported data.

Milestone 6: Conclusion

The Import Data using Transform Map project shows how ServiceNow can move data from an Excel spreadsheet into a target table in a controlled, repeatable way. A Data Source loads the file into a staging table, and a Transform Map with field maps, a coalesce rule and an optional script converts the data into clean records, updating existing ones and inserting new ones. Transform History provides a clear audit trail for verification and error correction. The approach removes manual entry, reduces duplicates and improves data quality. It can be extended by importing from other sources such as REST APIs, JDBC or LDAP, scheduling automatic imports, and adding validation and notification workflows for failed rows.